

Domain	Code	Service group	Smart ready service	Functionality level 0 (as non-smart default)	Functionality level 1	Functionality level 2	Functionality level 3	Functionality level 4	part of the method A: 1 - YES; 0 - NO	part of the method B: 1 - YES; 0 - NO	part of the custom services list?: 1 - YES; 0 - NO	Preconditions / Dependency on other services or building types	TRIAGE: 1 - This service affects maximum obtainable score, even if service is not applicable in this building; 0 - This service does not affect maximum obtainable score when not present in building
0	0	0	0	0	0	0	0	0	part of the method B: 1 - YES; 0 - NO			0	
0	0	0	0	0	0	0	0	0	part of the custom services list?: 1 - YES; 0 - NO			0	
Heating	H-1a	Heat control - demand side	Heat emission control	No automatic control	Central automatic control (e.g. central thermostat)	Individual room control (e.g. thermostatic valves, or electronic controller)	Individual room control with communication between controllers and to BACS	Individual room control with communication and occupancy detection	✔ 1	✔ 1		Triage: not relevant in case of TABS.	✘ 0
Heating	H-1b	Heat control - demand side	Emission control for TABS (heating mode)	No automatic control	Central automatic control	Advanced central automatic control	Advanced central automatic control with intermittent operation and/or room temperature feedback control		✘ 0	✔ 1		Triage: only relevant in case of TABS. Mostly restricted to non-residential buildings	✘ 0
Heating	H-1c	Heat control - demand side	Control of distribution fluid temperature (supply or return air flow or water flow) - Similar function can be applied to the control of direct electric heating networks	No automatic control	Outside temperature compensated control	Demand based control			✔ 1	✔ 1		Not applicable in case of individual heaters (e.g. stoves)	✘ 0
Heating	H-1d	Heat control - demand side	Control of distribution pumps in networks	No automatic control	On off control	Multi-Stage control	Variable speed pump control (pump unit (internal) estimations)	Variable speed pump control (external demand signal)	✘ 0	✔ 1		Only applicable for hydronic heating systems	✘ 0
Heating	H-1f	Heat control - demand side	Thermal Energy Storage (TES) for building heating (excluding TABS)	Continuous storage operation	Time-scheduled storage operation	Load prediction based storage operation	Heat storage capable of flexible control through grid signals (e.g. DSM)		✘ 0	✔ 1		Only applicable in case thermal energy storage is present.	✘ 0
Heating	H-2a	Control heat production facilities	Heat generator control (all except heat pumps)	Constant temperature control	Variable temperature control depending on outdoor temperature	Variable temperature control depending on the load (e.g. depending on supply water temperature set point)			✔ 1	✔ 1		Only applicable in case of combustion heater or district heating	✘ 0
Heating	H-2b	Control heat production facilities	Heat generator control (for heat pumps)	On/Off-control of heat generator	Multi-stage control of heat generator capacity depending on the load or demand (e.g. on/off of several compressors)	Variable control of heat generator capacity depending on the load or demand (e.g. hot gas bypass, inverter frequency control)	Variable control of heat generator capacity depending on the load AND external signals from grid		✔ 1	✔ 1		Only applicable in case of heat pumps	✘ 0
Heating	H-2d	Control heat production facilities	Sequencing in case of different heat generators	Priorities only based on running time	Control according to fixed priority list: e.g. based on rated energy efficiency	Control according to dynamic priority list (based on current energy efficiency, carbon emissions and capacity of generators, e.g. solar, geothermal heat, cogeneration plant, fossil fuels)	Control according to dynamic priority list (based on current AND predicted load, energy efficiency, carbon emissions and capacity of generators)	Control according to dynamic priority list (based on current AND predicted load, energy efficiency, carbon emissions, capacity of generators AND external signals from grid)	✘ 0	✔ 1		Only applicable in case of multiple heat generators, mostly restricted to large buildings	✘ 0
Heating	H-3	Information to occupants and facility managers	Report information regarding heating system performance	None	Central or remote reporting of current performance KPIs (e.g. temperatures, submetering energy usage)	Central or remote reporting of current performance KPIs and historical data	Central or remote reporting of performance evaluation including forecasting and/or benchmarking	Central or remote reporting of performance evaluation including forecasting and/or benchmarking	✔ 1	✔ 1		0	✔ 1
Heating	H-4	Flexibility and grid interaction	Flexibility and grid interaction	No automatic control	Scheduled operation of heating system	Self-learning optimal control of heating system	Heating system capable of flexible control through grid signals (e.g. DSM)	Optimized control of heating system based on local predictions and grid signals (e.g. through model predictive control)	✘ 0	✔ 1		The inspectability of the nature of the control algorithm would need to be facilitated	✔ 1
Domestic hot water	DHW-1a	Control DHW production facilities	Control of DHW storage charging (with direct electric heating or integrated electric heat pump)	Automatic control on / off	Automatic control on / off and scheduled charging enable	Automatic control on / off and scheduled charging enable and multi-sensor storage management	Automatic charging control based on local availability of renewables or information from electricity grid (DR, DSM)		✔ 1	✔ 1		Only applicable in case of DHW storage with electric heating	✘ 0
Domestic hot water	DHW-1b	Control DHW production facilities	Control of DHW storage charging (using hot water generation)	Automatic control on / off	Automatic control on / off and scheduled charging enable	Automatic on/off control, scheduled charging enable and demand-based supply temperature control or multi-sensor storage management	DHW production system capable of automatic charging control based on external signals (e.g. from district heating grid)		✔ 1	✔ 1		Only applicable in case of DHW storage with non-electrical heat generation	✘ 0
Domestic hot water	DHW-1d	Control DHW production facilities	Control of DHW storage charging (with solar collector and supplementary heat generation)	Manual selected control of solar energy or heat generation	Automatic control of solar storage charge (Prio. 1) and supplementary storage charge	Automatic control of solar storage charge (Prio. 1) and supplementary storage charge and demand-oriented supply or multi-sensor storage management	Automatic control of solar storage charge (Prio. 1) and supplementary storage charge, demand-oriented supply and return temperature control and multi-sensor storage management		✘ 0	✔ 1		Only applicable in case of DHW storage with solar collector	✘ 0
Domestic hot water	DHW-2b	Control DHW production facilities	Sequencing in case of different DHW generators	Priorities only based on running time	Control according to fixed priority list: e.g. based on rated energy efficiency	Control according to dynamic priority list (based on current energy efficiency, carbon emissions and capacity of generators, e.g. solar, geothermal heat, cogeneration plant, fossil fuels)	Control according to dynamic priority list (based on current AND predicted load, energy efficiency, carbon emissions and capacity of generators)	Control according to dynamic priority list (based on current AND predicted load, energy efficiency, carbon emissions, capacity of generators AND external signals from grid)	✘ 0	✔ 1		Only applicable in case of multiple heat generators, mostly restricted to large buildings	✘ 0

Domestic hot water	DHW-3	Information to occupants and facility managers	Report information regarding domestic hot water performance	None	Indication of actual values (e.g. temperatures, submetering energy usage)	Actual values and historical data	Performance evaluation including forecasting and/or benchmarking	Performance evaluation including forecasting and/or benchmarking; also including predictive management and fault detection	✔ 1	✔ 1		0	✔ 1
Cooling	C-1a	Cooling control - demand side	Cooling emission control	No automatic control	Central automatic control	Individual room control	Individual room control with communication between controllers and to BACS	Individual room control with communication and occupancy detection	✔ 1	✔ 1		Only applicable in case mechanical cooling systems are present	✘ 0
Cooling	C-1b	Cooling control - demand side	Emission control for TABS (cooling mode)	No automatic control	Central automatic control	Advanced central automatic control	Advanced central automatic control with intermittent operation and/or room temperature feedback control		✘ 0	✔ 1		Only applicable in case mechanical cooling systems based on TABS are present	✘ 0
Cooling	C-1c	Cooling control - demand side	Control of distribution network chilled water temperature (supply or return)	Constant temperature control	Outside temperature compensated control	Demand based control			✘ 0	✔ 1		Only applicable in case mechanical cooling systems with hydronic distribution system are present	✘ 0
Cooling	C-1d	Cooling control - demand side	Control of distribution pumps in networks	No automatic control	On off control	Multi-Stage control	Variable speed pump control (pump unit (internal) estimations)	Variable speed pump control (external demand signal)	✘ 0	✔ 1		Only applicable in case mechanical cooling systems with hydronic distribution system are present	✘ 0
Cooling	C-1f	Cooling control - demand side	Interlock: avoiding simultaneous heating and cooling in the same room	No interlock	Partial interlock (minimising risk of simultaneous heating and cooling e.g. by sliding setpoints)	Total interlock (control system ensures no simultaneous heating and cooling can take place)			✘ 0	✔ 1		Only applicable in case mechanical cooling systems are present	✔ 1
Cooling	C-1g	Cooling control - demand side	Control of Thermal Energy Storage (TES) operation	Continuous storage operation	Time-scheduled storage operation	Load prediction based storage operation	Cold storage capable of flexible control through grid signals (e.g. DSM)		✘ 0	✔ 1		Only applicable in case mechanical cooling systems are present and include TES systems	✘ 0
Cooling	C-2a	Control cooling production facilities	Generator control for cooling	On/Off-control of cooling production	Multi-stage control of cooling production capacity depending on the load or demand (e.g. on/off of several compressors)	Variable control of cooling production capacity depending on the load or demand (e.g. hot gas bypass, inverter frequency control)	Variable control of cooling production capacity depending on the load AND external signals from grid		✔ 1	✔ 1		Only applicable in case mechanical cooling systems are present	✔ 1
Cooling	C-2b	Control cooling production facilities	Sequencing of different cooling generators	Priorities only based on running times	Fixed sequencing based on loads only: e.g. depending on the generators characteristics such as absorption chiller vs. centrifugal chiller	Dynamic priorities based on generator efficiency and characteristics (e.g. availability of free cooling)	Load prediction based sequencing: the sequence is based on e.g. COP and available power of a device and the predicted required power	Sequencing based on dynamic priority list, including external signals from grid	✘ 0	✔ 1		Only applicable in case multiple mechanical cooling systems are present	✘ 0
Cooling	C-3	Information to occupants and facility managers	Report information regarding cooling system performance	None	Central or remote reporting of current performance KPIs (e.g. temperatures, submetering energy usage)	Central or remote reporting of current performance KPIs and historical data	Central or remote reporting of performance evaluation including forecasting and/or benchmarking	Central or remote reporting of performance evaluation including forecasting and/or benchmarking; also including predictive management and fault detection	✔ 1	✔ 1		Only applicable in case mechanical cooling systems are present	✔ 1
Cooling	C-4	Flexibility and grid interaction	Flexibility and grid interaction	No automatic control	Scheduled operation of cooling system	Self-learning optimal control of cooling system	Cooling system capable of flexible control through grid signals (e.g. DSM)	Optimized control of cooling system based on local predictions and grid signals (e.g. through model predictive control)	✔ 1	✔ 1		The inspectability of the nature of the control algorithm would need to be facilitated	✔ 1
Ventilation	V-1a	Air flow control	Supply air flow control at the room level	No ventilation system or manual control	Clock control	Occupancy detection control	Central Demand Control based on air quality sensors (CO2, VOC, humidity, ...)	Local Demand Control based on air quality sensors (CO2, VOC,...) with local flow from/to the zone regulated by dampers	✔ 1	✔ 1		Always to be assessed	✔ 1
Ventilation	V-1c	Air flow control	Air flow or pressure control at the air handler level	No automatic control: Continuously supplies of air flow for a maximum load of all rooms	On off time control: Continuously supplies of air flow for a maximum load of all rooms during nominal occupancy time	Multi-stage control: To reduce the auxiliary energy demand of the fan	Automatic flow or pressure control without pressure reset: Load dependent supplies of air flow for the demand of all connected rooms.	Automatic flow or pressure control with pressure reset: Load dependent supplies of air flow for the demand of all connected rooms (for variable air volume systems with VFD).	✘ 0	✔ 1		Only in case of mechanical ventilation	✘ 0
Ventilation	V-2c	Air temperature control	Heat recovery control: prevention of overheating	Without overheating control	Modulate or bypass heat recovery based on sensors in air exhaust	Modulate or bypass heat recovery based on multiple room temperature sensors or predictive control			✘ 0	✔ 1		Only in case of mechanical ventilation with heat recovery	✘ 0
Ventilation	V-2d	Air temperature control	Supply air temperature control at the air handling unit level	No automatic control	Constant setpoint: A control loop enables to control the supply air temperature, the setpoint is constant and can only be modified by a manual action	Variable set point with outdoor temperature compensation	Variable set point with load dependant compensation. A control loop enables to control the supply air temperature. The setpoint is defined as a function of the loads in the room		✘ 0	✔ 1		Only in case of mechanical ventilation which supplies heating	✘ 0
Ventilation	V-3	Free cooling	Free cooling with mechanical ventilation system	No automatic control	Night cooling	Free cooling: air flows modulated during all periods of time to minimize the amount of mechanical cooling	H,x- directed control: The amount of outside air and recirculation air are modulated during all periods of time to minimize the amount of mechanical cooling. Calculation is performed on the basis of temperatures and humidity (enthalpy).		✘ 0	✔ 1		Only in case of mechanical or hybrid ventilation	✘ 0
Ventilation	V-6	Feedback - Reporting information	Reporting information regarding IAQ	None	Air quality sensors (e.g. CO2) and real time autonomous monitoring	Real time monitoring & historical information of IAQ available to occupants	Real time monitoring & historical information of IAQ available to occupants + warning on maintenance needs or occupant actions (e.g. window opening)		✔ 1	✔ 1		Always to be assessed	✔ 1
Lighting	L-1a	Artificial lighting control	Occupancy control for indoor lighting	Manual on/off switch	Manual on/off switch + additional sweeping extinction signal	Automatic detection (auto on / dimmed or auto off)	Automatic detection (manual on / dimmed or auto off)		✔ 1	✔ 1		Always to be assessed	✔ 1
Lighting	L-2	Control artificial lighting power based on daylight levels	Control artificial lighting power based on daylight levels	Manual (central)	Manual (per room / zone)	Automatic switching	Automatic dimming	Automatic dimming including scene-based light control (during time intervals, dynamic and adapted lighting scenes are set, for example, in terms of illuminance level, different correlated colour temperature (CCT)	✘ 0	✔ 1		Always to be assessed	✔ 1
Dynamic building envelope	DE-1	Window control	Window solar shading control	No sun shading or only manual operation	Motorized operation with manual control	Motorized operation with automatic control based on sensor data	Combined light/blind/HVAC control	Predictive blind control (e.g. based on weather forecast)	✔ 1	✔ 1		Only applicable in case movable shades, screens or blinds are present	✘ 0

Dynamic building envelope	DE-2	Window control	Window open/closed control, combined with HVAC system	Manual operation or only fixed windows	Open/closed detection to shut down heating or cooling systems	Level 1 + Automised mechanical window opening based on room sensor data	Level 2 + Centralized coordination of operable windows, e.g. to control free natural night cooling		✗	0	✓	1		0	✓	1
Dynamic building envelope	DE-4	Feedback - Reporting information	Reporting information regarding performance of dynamic building envelope systems	No reporting	Position of each product & fault detection	Position of each product, fault detection & predictive maintenance	Position of each product, fault detection, predictive maintenance, real-time sensor data (wind, lux, temperature...)	Position of each product, fault detection, predictive maintenance, real-time & historical sensor data (wind, lux, temperature...)	✓	1	✓	1		Only applicable in case movable shades, screens or blinds are present	✗	0
Electricity	E-2	Feedback - Reporting information	Reporting information regarding local electricity generation	None	Current generation data available	Actual values and historical data	Performance evaluation including forecasting and/or benchmarking	Performance evaluation including forecasting and/or benchmarking; also including predictive management and fault detection	✓	1	✓	1		Only applicable in case of local energy generation	✗	0
Electricity	E-3	DER - Storage	Storage of (locally generated) electricity	None	On site storage of electricity (e.g. electric battery)	On site storage of energy (e.g. electric battery or thermal storage) with controller based on grid signals	On site storage of energy (e.g. electric battery or thermal storage) with controller optimising the use of locally generated electricity	On site storage of energy (e.g. electric battery or thermal storage) with controller optimising the use of locally generated electricity and possibility to feed back into the grid	✓	1	✓	1		Only applicable in case of local energy generation	✗	0
Electricity	E-4	DER- Optimization	Optimizing self-consumption of locally generated electricity	None	Scheduling electricity consumption (plug loads, white goods, etc.)	Automated management of local electricity consumption based on current renewable energy availability	Automated management of local electricity consumption based on current and predicted energy needs and renewable energy availability		✗	0	✓	1		Only applicable in case of local energy generation	✗	0
Electricity	E-5	DER - Generation Control	Control of combined heat and power plant (CHP)	CHP control based on scheduled runtime management and/or current heat energy demand	CHP runtime control influenced by the fluctuating availability of RES; overproduction will be fed into the grid	CHP runtime control influenced by the fluctuating availability of RES and grid signals; dynamic charging and runtime control to optimise self-consumption of renewables			✗	0	✓	1		Only applicable in case of CHP	✗	0
Electricity	E-8	DSM- Storage	Support of (micro)grid operation modes	None	Automated management of (building-level) electricity consumption based on grid signals	Automated management of (building-level) electricity consumption and electricity supply to neighbouring buildings (microgrid) or grid	Automated management of (building-level) electricity consumption and supply, with potential to continue limited off-grid operation (island mode)		✗	0	✓	1		Only applicable in case of local energy storage	✗	0
Electricity	E-11	Feedback - Reporting information	Reporting information regarding energy storage	None	Current state of charge (SOC) data available	Actual values and historical data	Performance evaluation including forecasting and/or benchmarking	Performance evaluation including forecasting and/or benchmarking; also including predictive management and fault detection	✓	1	✓	1		Only applicable in case of local energy storage	✗	0
Electricity	E-12	Feedback - Reporting information	Reporting information regarding electricity consumption	None	reporting on current electricity consumption on building level	real-time feedback or benchmarking on building level	real-time feedback or benchmarking on appliance level	real-time feedback or benchmarking on appliance level with automated personalized recommendations	✓	1	✓	1		Always to be assessed	✓	1
Electric vehicle charging	EV-15	EV Charging	EV Charging Capacity	not present	ducting (or simple power plug) available	0-9% of parking spaces has recharging points	10-50% of parking spaces has recharging point	>50% of parking spaces has recharging point	✓	1	✓	1		Only to be assessed if parking spots available on site	✗	0
Electric vehicle charging	EV-16	EV Charging - Grid	EV Charging Grid balancing	Not present (uncontrolled charging)	1-way controlled charging (e.g. including desired departure time and grid signals for optimization)	2-way controlled charging (e.g. including desired departure time and grid signals for optimization)			✓	1	✓	1		Only to be assessed if EV charging available on site	✗	0
Electric vehicle charging	EV-17	EV Charging - connectivity	EV charging information and connectivity	No information available	Reporting information on EV charging status to occupant	Reporting information on EV charging status to occupant AND automatic identification and authorization of the driver to the charging station (ISO 15118 compliant)			✓	1	✓	1		Only to be assessed if EV charging available on site	✗	0
Monitoring and control	MC-3	HVAC interaction control	Run time management of HVAC systems	Manual setting	Runtime setting of heating and cooling plants following a predefined time schedule	Heating and cooling plant on/off control based on building loads	Heating and cooling plant on/off control based on predictive control or grid signals		✗	0	✓	1		0	✓	1
Monitoring and control	MC-4	Fault detection	Detecting faults of technical building systems and providing support to the diagnosis of these faults	No central indication of detected faults and alarms	With central indication of detected faults and alarms for at least 2 relevant TBS	With central indication of detected faults and alarms for all relevant TBS	With central indication of detected faults and alarms for all relevant TBS, including diagnosing functions		✗	0	✓	1		0	✓	1
Monitoring and control	MC-9	TBS interaction control	Occupancy detection: connected services	None	Occupancy detection for individual functions, e.g. lighting	Centralised occupant detection which feeds in to several TBS such as lighting and heating			✗	0	✓	1		0	✓	1
Monitoring and control	MC-13	Feedback - Reporting information	Central reporting of TBS performance and energy use	None	Central or remote reporting of realtime energy use per energy carrier	Central or remote reporting of realtime energy use per energy carrier, combining TBS of at least 2 domains in one interface	Central or remote reporting of realtime energy use per energy carrier, combining TBS of all main domains in one interface		✓	1	✓	1		0	✓	1
Monitoring and control	MC-25	Smart Grid Integration	Smart Grid Integration	None - No harmonization between grid and TBS; building is operated independently from the grid load	Demand side management possible for (some) individual TBS, but not coordinated over various domains	Coordinated demand side management of multiple TBS			✓	1	✓	1		The inspectability of the nature of the control algorithm would need to be facilitated for level 2. Service 7.5 in EN15232-1-17. Average impacts derived from multiple simulations to produce BACS factors in EN15232.	✓	1
Monitoring and control	MC-28	Feedback - Reporting information	Reporting information regarding demand side management performance and operation	None	Reporting information on current DSM status, including managed energy flows	Reporting information on current DSM status, including managed energy flows			✗	0	✓	1		0	✓	1
Monitoring and control	MC-29	Override control	Override of DSM control	No DSM control	DSM control without the possibility to override this control by the building user (occupant or facility manager)	Manual override and reactivation of DSM control by the building user	Scheduled override of DSM control (and reactivation) by the building user	Scheduled override of DSM control and reactivation with optimised control	✗	0	✓	1		0	✓	1

Monitoring and control	MC-30	Single platform that allows automated control & coordination between TBS + optimization of energy flow based on occupancy, weather and grid signals	Single platform that allows automated control & coordination between TBS + optimization of energy flow based on occupancy, weather and grid signals	None	Single platform that allows manual control of multiple TBS	Single platform that allows automated control & coordination between TBS	Single platform that allows automated control & coordination between TBS + optimization of energy flow based on occupancy, weather and grid signals		✓ 1	✓ 1		Always to be assessed	✓ 1
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